

JOHN ZHANG

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EDUCATION

UNIVERSITY OF TORONTO

Bachelor of Science

Major in Computer Science; Major in Statistical Science

Cumulative GPA: 3.89/4.0; Innis College Anniversary Scholarship 2025

Relevant Coursework: Machine Learning, Data Analysis, Statistics, Deep Learning, Data Structures & Algorithms

Toronto, ON

Expected Jan 2028

WORK EXPERIENCE

SUNNYBROOK RESEARCH INSTITUTE

Machine Learning Analyst

Toronto, ON

May 2025 – Present

- Engineered deep learning models (CNN, LSTM) to translate data between modern and legacy accelerometers, enabling reuse of historical actigraphy data in clinical research and contributing to an upcoming publication.
- Developed and evaluated machine learning models for Alzheimer's disease prediction using EEG and sleep-stage data, including feature engineering and subject-level validation.
- Designed and containerized a scalable sleep-wake classification pipeline with Docker for reproducible analysis of large-scale UK Biobank actigraphy data.

UNIVERSITY OF TORONTO AEROSPACE TEAM

Data Processing Engineer

Toronto, ON

Sep 2024 – Jan 2025

- Debugged and modernized legacy Python and SQL code while researching hyperspectral image correction methods, restoring satellite-data processing capabilities and improving image accuracy for scientific analysis.

PROJECTS

AUTOML AGENT

[Live Demo](#) | [GitHub Repo](#) | Jul 2026 – Present

- Architected a multi-agent LLM pipeline (LangGraph) that turns a natural-language ML request into a trained model, automating dataset analysis, experiment planning, model generation, and result reporting end-to-end.
- Built a retrieval-augmented research agent that embeds the request and dataset profile, retrieves relevant excerpts from a Qdrant vector store, and grounds the planning agent throughout the experimentation loop.
- Engineered a self-repairing experimentation agent that generates library-agnostic Python training code, installs dependencies in an isolated environment, and automatically diagnoses and fixes runtime failures.
- Built production ML lifecycle tooling; live prediction serving, drift monitoring (Evidently) and a one-click retraining flow; and a LangSmith evaluation suite covering all agents, integrated into CI to catch regressions.
- Deployed the full system on Google Cloud Platform; FastAPI backend on Cloud Run, persistence via Cloud Storage and Firestore, pipeline observability via Cloud Monitoring, and a React frontend on Firebase Hosting.

COST-AWARE TOOL-USE LEARNING FOR LLM

[GitHub Repo](#) | Sep 2026 – Present

- Built a modular tool-use agent framework in Python (Tool/Registry/Executor/Environment/Agent) with 26+ tools, using Pydantic for structured, type-safe state throughout.
- Built an LLM agent selecting tool calls via structured JSON output, generating teacher trajectories distilled into a fine-tuned model, evaluated via an LLM-as-judge harness scoring task success, tool accuracy, cost, and latency.
- Fine-tuned Qwen2.5 on agent tool usage trajectories using LoRA and QLoRA (Hugging Face PEFT and Transformers), with custom tokenization, 4-bit quantization, and hyperparameter tuning.

TECHNICAL SKILLS

Machine Learning: PyTorch, TensorFlow, Keras, Scikit-Learn, Deep Learning, Machine Learning, Evidently, Hugging Face (Transformers, PEFT), LoRA & QLoRA Fine-Tuning, TensorBoard

Engineering: Docker, Git, FastAPI, LangGraph, LangSmith, RAG, Qdrant, CI/CD (GitHub Actions), LLM

Data/Cloud: NumPy, Pandas, SciPy, Google Cloud Platform, SQL, Signal Processing